

Tonal Languages

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2023-04-28

Modeling

Probit Model

```
glm_tone_probit <- glm(data = tones_any,
                      formula = is_tonal ~ dist_equator,
                      family = binomial(link = probit))
summary(glm_tone_probit)

##
## Call:
## glm(formula = is_tonal ~ dist_equator, family = binomial(link = probit),
##      data = tones_any)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3651  -1.0940  -0.6496   1.1409   2.1863
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.269185   0.090443   2.976  0.00292 **
## dist_equator -0.023880   0.003675  -6.497 8.17e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 716.15  on 526  degrees of freedom
## Residual deviance: 669.98  on 525  degrees of freedom
## AIC: 673.98
##
## Number of Fisher Scoring iterations: 3
```

Logit Model

```
glm_tone_logit <- glm(data = tones_any,
                     formula = is_tonal ~ dist_equator,
                     family = binomial(link = logit))
summary(glm_tone_logit)

##
## Call:
```

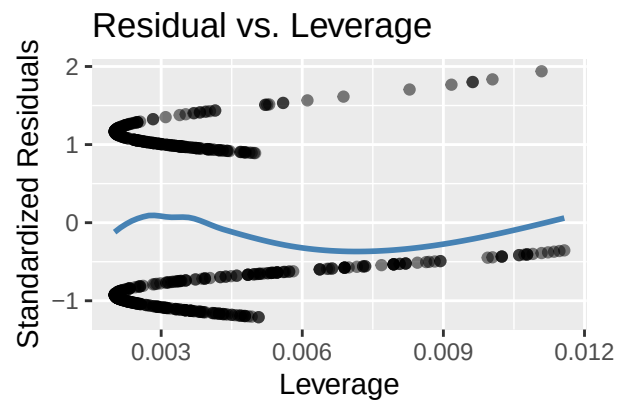
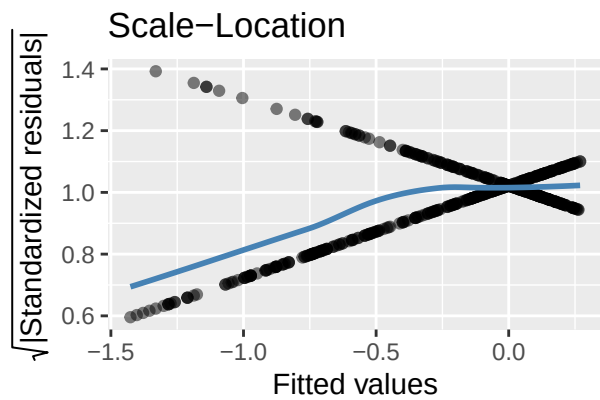
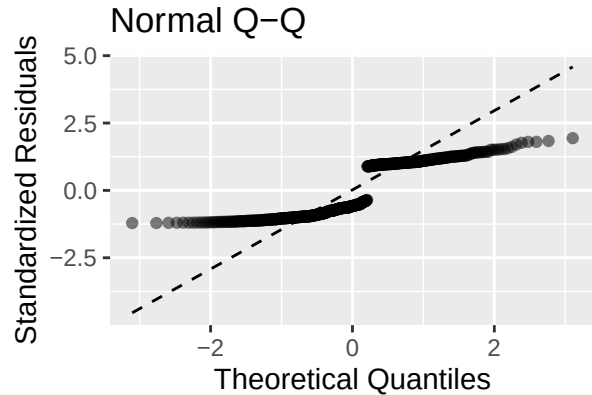
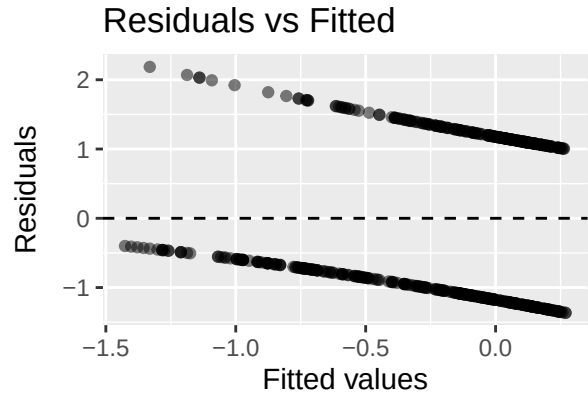
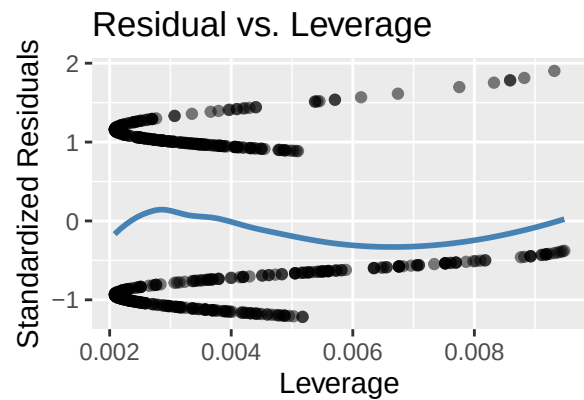
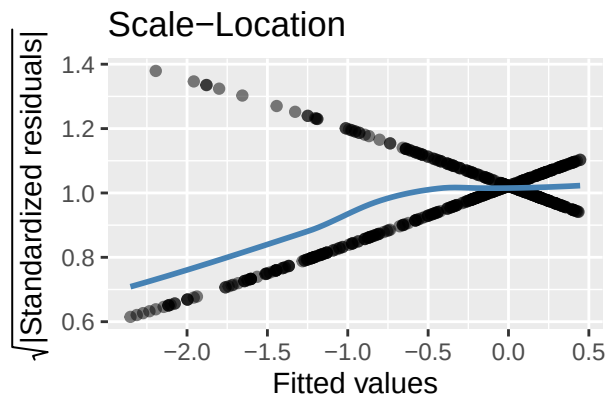
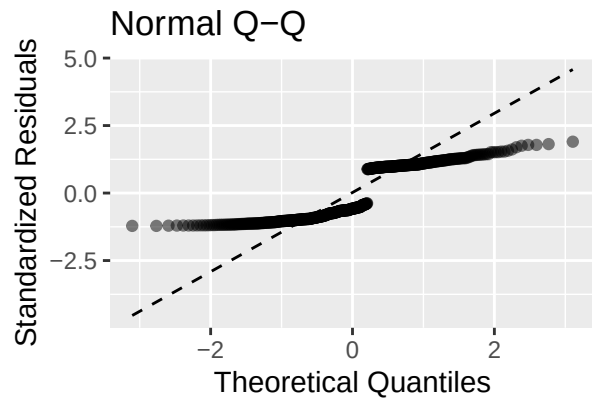
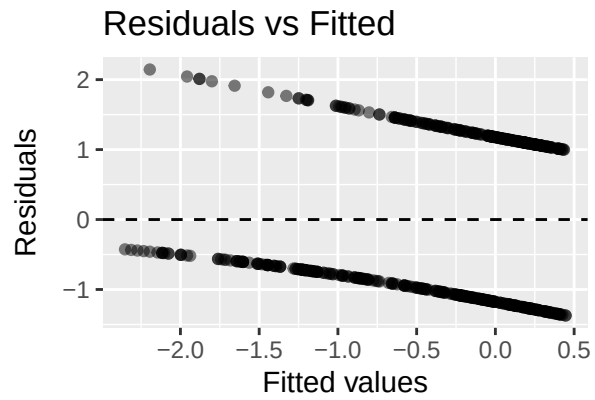
```
## glm(formula = is_tonal ~ dist_equator, family = binomial(link = logit),
##     data = tones_any)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3716  -1.0919  -0.6504   1.1390   2.1448
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.445725   0.147442   3.023   0.0025 **
## dist_equator -0.039406   0.006311  -6.244 4.26e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 716.15  on 526  degrees of freedom
## Residual deviance: 670.00  on 525  degrees of freedom
## AIC: 674
##
## Number of Fisher Scoring iterations: 3
```

Transformed Probit Model

```
glm_tone_probit <- glm(data = tones_any,
                      formula = I(is_tonal^2) ~ dist_equator,
                      family = binomial(link = probit))
summary(glm_tone_probit)

##
## Call:
## glm(formula = I(is_tonal^2) ~ dist_equator, family = binomial(link = probit),
##     data = tones_any)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3651  -1.0940  -0.6496   1.1409   2.1863
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   0.269185   0.090443   2.976 0.00292 **
## dist_equator -0.023880   0.003675  -6.497 8.17e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 716.15  on 526  degrees of freedom
## Residual deviance: 669.98  on 525  degrees of freedom
## AIC: 673.98
##
## Number of Fisher Scoring iterations: 3
```

Residuals



Dispersion Estimates

```
glm_tone_probit <- glm(data = tones_any,
                      formula = I(is_tonal) ~ dist_equator,
                      family = quasibinomial(link = probit))
summary(glm_tone_probit)

##
## Call:
## glm(formula = I(is_tonal) ~ dist_equator, family = quasibinomial(link = probit),
##      data = tones_any)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.3651  -1.0940  -0.6496   1.1409   2.1863
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.269185   0.090680   2.969  0.00313 **
## dist_equator -0.023880   0.003685  -6.481 2.11e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for quasibinomial family taken to be 1.005243)
##
##      Null deviance: 716.15  on 526  degrees of freedom
## Residual deviance: 669.98  on 525  degrees of freedom
## AIC: NA
##
## Number of Fisher Scoring iterations: 3
```